

1. Different Charts Supported in Power BI

Power BI supports a wide range of **visuals (charts)** for data visualization. Commonly used ones include:

Chart Type	Purpose / Usage
Column Chart	Compare values across categories (e.g., sales by region).
Bar Chart	Same as column chart, but horizontal — better for long category names.
Line Chart	Show trends over time (e.g., monthly revenue trend).
Area Chart	Similar to line chart, but with shaded area — shows magnitude of change.
Pie / Donut Chart	Show proportions or percentage contribution to a total.
Tree Map	Display hierarchical data using nested rectangles — good for part-to-whole analysis.
Scatter Chart	Show relationships between two numerical variables (e.g., profit vs. sales).
Waterfall Chart	Show how an initial value increases or decreases through a series of changes (e.g., profit breakdown).
Funnel Chart	Show stages in a process (e.g., sales pipeline).
Gauge / KPI Card	Show performance against a target.
Map / Filled Map	Show geographical data (e.g., sales by country or region).
Combo Chart	Combine two chart types (e.g., line + column) to compare two measures with different scales.

2. When to Use Which Chart (With Example)

Chart Type	When to Use	Example
Column Chart	When you want to compare categories.	Compare total sales by product category .
Line Chart	When analyzing trends over time.	Track monthly revenue growth for the year.
Pie Chart	When showing parts of a whole (limit to <5 categories).	Show market share percentage of each product.
Tree Map	When you have hierarchical data and want to compare proportions.	Show sales by region and product category in one view.
Scatter Chart	When you want to identify correlation between measures.	Analyze sales vs. profit to see which products perform better.
Waterfall Chart	When analyzing sequential changes in value.	Show how revenue leads to final profit after costs and taxes.

Chart Type	When to Use	Example
Funnel Chart	When showing step-wise process drop-off.	Visualize lead conversion stages in a sales funnel.
Gauge Chart	When monitoring KPIs against a target.	Show if monthly sales reached target .
Map Chart	When showing location-based insights.	Display sales by country using geographic visualization.

3. Difference Between Matrix Visual and Table Visual

Feature	Table Visual	Matrix Visual
Structure	Displays data in a flat, tabular format (like Excel table).	Displays data in a pivot-table style with rows and columns.
Data Layout	Single-level rows and columns.	Supports multiple levels (hierarchies) in rows and columns.
Aggregation	Shows detailed data only.	Can aggregate and summarize values dynamically.
Drill-down Support	No	Yes (you can expand/collapse hierarchies).
Use Case	When you need a simple tabular report.	When you want summarized, pivot-style analysis.

Example:

- **Table Visual** → Show each **transaction detail** (Date, Customer, Product, Sales).
- **Matrix Visual** → Show **Total Sales by Region and Product Category**, with the ability to **expand/collapse** categories.